Long-term toxicity of a Roundup herbicide and Roundup-tolerant GM maize

Review summary (based on Séralini et al., 2012; republished 2014)

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Study question & rationale

Core question

Does 2-year (lifetime) exposure to:

- GM maize NK603 (Roundup-tolerant)
- Roundup formulation in drinking water

…or both lead to adverse outcomes in rats?

Why 2 years?

- Many regulatory feeding studies are ~90 days
- Authors argue late-onset tumors/organ disease may appear after 90 days
- Goal: detect long-term signals across mortality, tumors, organs, and biomarkers

Model organism: rats

virgin albino Sprague–Dawley rats used by authors

Experimental design (who, how many, how allocated)

Animals

- Virgin albino Sprague–Dawley rats, ~5 weeks old at start
- Total n = 200 (100 ♂, 100 ♀)
- Each sex was split into 10 groups, with 10 rats per group.
- Followed for 2 years (lifetime)

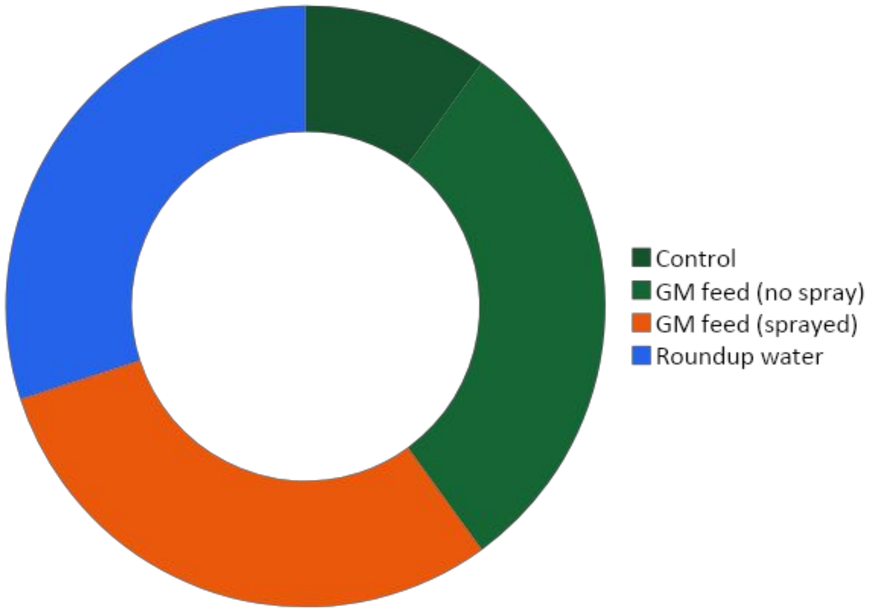
Exposure logic

Two main routes were separated:

- GM maize in feed (NK603)

Roundup formulation in drinking water plus a distinction for GM maize: grown with vs without Roundup spraying.

10 groups per sex (structure)



Each group: 10 rats/sex → outcomes like tumors/mortality can be noisy.

Exposure groups (per sex)

Control	GM maize (NK603) no Roundup spray	GM maize (NK603) Roundup-sprayed crop	Roundup in drinking water
• Non-GM maize diet • Plain water	• 11% NK603 + water • 22% NK603 + water • 33% NK603 + water	• 11% NK603 (sprayed) + water • 22% NK603 (sprayed) + water • 33% NK603 (sprayed) + water	• Control diet + 1.1 $\times 10^{-8}$% Roundup • Control diet + 0.09% Roundup • Control diet + 0.5% Roundup

Reported results (1/2): mortality & tumors

Mortality

- Controls lived longest in authors’ plots
- Treated groups often showed earlier/higher mortality
- Authors highlight especially strong effects in females (claimed up to 2–3× deaths in some arms)

Tumors (emphasis: females)

- Large palpable tumors reported more often in treated groups
- Female outcomes stressed: mammary/pituitary tumor burden
- Effects described as not dose-linear (non-monotonic patterns suggested)

Interpretive caveats

- n=10/sex/group → tumor & survival rates can swing with 1–2 animals
- Many outcomes × many groups → high multiple-comparison burden unless primary endpoints are pre-specified which is not
- Visual/descriptive emphasis can overstate certainty without formal inference

Reported results (2/2): organs & biomarkers

Pathology signals (authors' framing)	Biochemistry & hormones
• Males: more frequent liver lesions & kidney disease reported across treatments • Organ weights and tissue exams used to support "hepatorenal toxicity" • Effects again described as not consistently dose-proportional	• Repeated blood/urine panels → many biomarkers • Authors emphasize multivariate analysis around month 15 • Reported shifts in kidney-related markers + sex-hormone balance

Method watch-outs (why biomarker claims are fragile here)

- Month-15 focus can introduce survivorship bias (healthiest animals remain)
- Small groups + multivariate modeling → overfitting risk
- Without a pre-declared analysis plan, it's hard to separate signal vs chance

Hypotheses (mostly implicit in the paper)

The paper does not clearly pre-specify formal H0/H1 hypothesis or a primary endpoint. Instead, the implied hypotheses below are tested across many groups and many outcomes, which later affects interpretability.

- Long-term toxicity: 2-year exposure to NK603 (grown with or without Roundup cultivation) and/or Roundup in water increases adverse outcomes (mortality, tumors, organ pathology, clinical chemistry).
- Sex-specific / endocrine-related: effects differ by sex and relate to hormonal disruption (females: mammary/pituitary focus; males: hepatorenal focus; hormone markers differ).
- Non-linear dose-response: effects may be thresholded or non-monotonic rather than dose-proportional.

❗ without a primary endpoint plan, the study effectively becomes a large multiple-comparisons exercise. ❗

Was the design appropriate for the hypotheses?

Strengths

- 2-year study with frequent monitoring matches the claim that 90-day studies may miss long-term effects.
- The 10-group structure aligns with separating “GM diet” vs “Roundup exposure route.”

Weaknesses

- Low Low power for key claims: only 10 rats per group (for each sex). For endpoints like tumors/mortality, this is typically too small to support strong causal claims (high noise, unstable rates).
- Many comparisons: 10 groups per sex + many biomarkers → high multiplicity burden unless the design is paired with a strict primary endpoint plan (not present).
- Potential repeatability/validity gaps: blinding and unit-of-analysis issues (e.g., 2 rats/cage) aren’t clearly addressed, which matters for reliability of pathology and tumor outcomes.

Were the analysis and conclusion appropriate?

- Mortality/tumor evidence is largely descriptive in figures (curves/histograms) and strong verbal claims are made (e.g., "2–3 times more deaths" in females).
- Biochemistry relies heavily on month-15 OPLS-DA (chosen because most animals are still alive), which increases risk of survivorship bias and model overfitting in small samples.
- Conclusions vs evidence: The paper's language implies fairly definitive harm at low exposures and suggests mechanisms (endocrine disruption). Given the design/power and analysis limitations, the conclusions read stronger than warranted by the reported evidence.

General review: Does the paper give a good account of the research conducted? Why (not)?

- As a hypothesis-generating long-term study, it raises signals worth further testing.
- As a confirmatory demonstration of causal effects on mortality/tumors, it does not meet common scientific-quality expectations due to low power, many comparisons, and limited inferential transparency.

Analysis, conclusions, and overall judgment

Overall, Séralini et al. (2012) reads more as an exploratory, hypothesis-generating lifetime feeding study than a confirmatory test: while the 2-year duration and broad monitoring are aligned with the authors' goal of detecting long-term effects, the paper does not clearly pre-specify primary hypotheses/endpoints or a confirmatory analysis plan, uses small group sizes for outcomes as variable as tumors and mortality, and relies heavily on visually persuasive but inferentially weak figures, plus multivariate biomarker modeling at a survivorship-affected timepoint; taken together, these features elevate risks of low power, multiple-comparisons false positives, and bias, so the strongest causal claims are not well supported by the design and analysis.